Chapter 9. Gaussian Linear SEMs and Directed Relations

Graphical models, Lecture 2 (DAGs, chain/fork/collider, d-separation, faithfulness, Markov equivalence, CPDAG)

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1 From Undirected to Directed Graphs

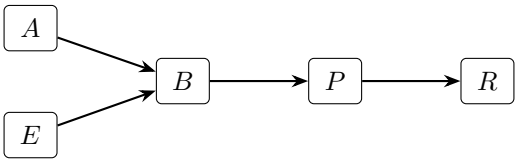
Lecture 1: conditional-dependence graph	Lecture 2: directed graph
$X_j - X_k \iff X_j \not\perp\!\!\!\perp X_k \mid X_{-\{j,k\}}$ finds relations that remain after accounting for the others edges but no directions	$X_k \to X_j$ expresses each arrow by a generating equation per variable reads the conditional independences the graph creates; checks which directions data cannot distinguish

The key question: **how far can the direction of relations be inferred from Gaussian observational data?**

2 A Gaussian Linear SEM Is a Bundle of Regressions on a DAG

2.1 The health example as equations

Consider the generating relations $A \to B \leftarrow E$ and $B \to P \to R$:



$$A = \epsilon_A, \quad E = \epsilon_E, \quad B = \beta_{AB}A + \beta_{EB}E + \epsilon_B, \quad P = \beta_{BP}B + \epsilon_P, \quad R = \beta_{PR}P + \epsilon_R,$$
$$\epsilon_j \overset{\text{ind}}{\sim} N(0, \sigma_j^2).$$

Each variable is a linear combination of its parents plus its own error; with independent Gaussian errors the whole vector is multivariate Gaussian.

Definition: DAG and Gaussian linear SEM

A directed acyclic graph (DAG) G has directed edges and no directed cycle. With $\text{pa}(j)$ the parents of node j,

$$X_j = \sum_{k \in \text{pa}(j)} \beta_{kj} X_k + \epsilon_j, \qquad \epsilon_1, \ldots, \epsilon_p \text{ independent}, \ \epsilon_j \sim N(0, \sigma_j^2).$$

Variable	Parents	Local regression
A, E	none	$A = \epsilon_A$, $E = \epsilon_E$
B	A, E	$B = \beta_{AB}A + \beta_{EB}E + \epsilon_B$
P	B	$P = \beta_{BP}B + \epsilon_P$
R	P	$R = \beta_{PR}P + \epsilon_R$

For instance exercise E does not enter the equation of blood pressure P directly, but is related to P through the path $E \to B \to P$.

2.2 Matrix form and the implied covariance

Let B be the $p \times p$ matrix with $B_{jk} = \beta_{kj}$ (row j holds the coefficients of the equation of X_j) and $D = \mathrm{diag}(\sigma_1^2, \ldots, \sigma_p^2)$. Then $X = BX + \epsilon$, and because G is acyclic, B can be permuted to be strictly lower triangular, so $I - B$ is invertible:

$$X = (I - B)^{-1}\epsilon, \qquad \Sigma = (I - B)^{-1} D (I - B)^{-\top}, \qquad \Omega = \Sigma^{-1} = (I - B)^{\top} D^{-1} (I - B).$$

The density factorizes along the DAG (**Markov factorization**):

$$p(x_1, \ldots, x_p) = \prod_{j=1}^{p} p(x_j \mid x_{\mathrm{pa}(j)}).$$

Theorem: the GGM of an SEM is its moral graph

For $j \neq k$,

$$\Omega_{jk} = -\frac{B_{jk}}{\sigma_j^2} - \frac{B_{kj}}{\sigma_k^2} + \sum_{l \neq j,k} \frac{B_{lj} B_{lk}}{\sigma_l^2}.$$

Hence $\Omega_{jk} \neq 0$ only if j, k are adjacent in G or have a common child; generically (no cancellation) exactly then. The GGM of Ch. 8 is the **moral graph**: the skeleton plus edges "marrying" the parents of each child.

Proof [Supplement]. $\Omega_{jk} = \sum_l (I - B)_{lj}(I - B)_{lk}/\sigma_l^2$. The term $l = j$ gives $(1)(-B_{jk})/\sigma_j^2$, the term $l = k$ gives $(-B_{kj})(1)/\sigma_k^2$, and $l \neq j, k$ gives $B_{lj}B_{lk}/\sigma_l^2$, nonzero only if l is a child of both. ■

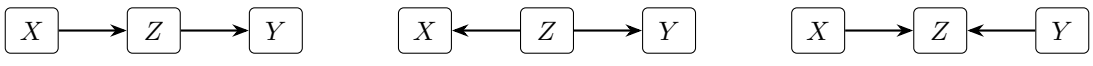
In the health example A and E are independent, yet the GGM contains $A - E$, because both are parents of B: conditioning on B (among all other variables) connects them. This is the first warning that undirected conditional relations are not causal edges.

3 Chains, Forks and Colliders

Motivation. Every path in a DAG is built from three local patterns. Chains and forks transmit association unless the middle variable is conditioned on; colliders block it unless conditioned on. These rules are the basis of d-separation and of all constraint-based discovery.

Structure	Graph	Marginally	Conditioning on Z
chain	$X \to Z \to Y$	X and Y related	$X \perp\!\!\!\perp Y \mid Z$
fork	$X \leftarrow Z \to Y$	X and Y related	$X \perp\!\!\!\perp Y \mid Z$
collider	$X \to Z \leftarrow Y$	$X \perp\!\!\!\perp Y$	X and Y related

These are the typical patterns observed when the Markov condition and faithfulness hold (§4). Conditioning **closes** chains and forks and **opens** colliders.



chain: open given Z: blocked | fork: open given Z: blocked | collider: blocked given Z: open

Figure 1: The three elementary paths and what conditioning on the middle node does to the flow of association between X and Y. Only the collider reverses the pattern, which is why v-structures are detectable from data. Conditioning on a *descendant* of a collider also opens it.

Theorem: the three patterns in a Gaussian SEM

Let all errors be independent with variances 1 and coefficients $a, b \neq 0$.

1. Chain $Z = aX + \epsilon_Z$, $Y = bZ + \epsilon_Y$: $\mathrm{Cov}(X, Y) = ab \neq 0$, $\rho_{XY \cdot Z} = 0$.
2. Fork $X = aZ + \epsilon_X$, $Y = bZ + \epsilon_Y$: $\mathrm{Cov}(X, Y) = ab \neq 0$, $\rho_{XY \cdot Z} = 0$.
3. Collider $Z = aX + bY + \epsilon_Z$ with $X \perp\!\!\!\perp Y$: $\mathrm{Cov}(X, Y) = 0$ but $\rho_{XY \cdot Z} = -\dfrac{ab}{\sqrt{(1 + b^2)(1 + a^2)}} \neq 0$.

Proof [Supplement]. (1) $Y = abX + b\epsilon_Z + \epsilon_Y$ so $\mathrm{Cov}(X, Y) = ab$; the residual of Y on Z is ϵ_Y, independent of (X, Z), so the partial correlation is 0.

(2) $\mathrm{Cov}(X, Y) = ab\mathrm{Var}Z$; residuals of X, Y on Z are ϵ_X, ϵ_Y, independent.

(3) $\mathrm{Cov}(X, Y) = 0$. With $\mathrm{Var}Z = a^2 + b^2 + 1$, $\mathrm{Cov}(X, Z) = a$, $\mathrm{Cov}(Y, Z) = b$, the residual covariance is $0 - ab/(a^2 + b^2 + 1)$ and the residual variances are $1 - a^2/(a^2 + b^2 + 1) = (1 + b^2)/(a^2 + b^2 + 1)$ and $(1 + a^2)/(a^2 + b^2 + 1)$; divide. ■

In a collider, telling us that the effect Z has a given value makes information about one cause change what we know about the other ("explaining away"): if BMI is high and age is low, exercise is probably low. **The change of conditional independence is the key information that distinguishes a collider from the other two structures.**

4 d-Separation, the Markov Condition and Faithfulness

Definition: d-separation

A path between X and Y is **blocked** by a set S if it contains (i) a non-collider (chain or fork node) that is in S, or (ii) a collider that is not in S and has no descendant in S. X and Y are d-separated by S, written $X \perp_G Y \mid S$, if every path between them is blocked.

Definition: causal Markov condition and faithfulness

Markov: $X \perp_G Y \mid S \Rightarrow X \perp\!\!\!\perp Y \mid S$ in the distribution (equivalent to the Markov factorization). **Faithfulness**: the converse, $X \perp\!\!\!\perp Y \mid S \Rightarrow X \perp_G Y \mid S$: every conditional independence in the distribution is implied by the graph.

Under both, the conditional independences of the data are *exactly* the d-separations of G, which is what structure learning (Ch. 10) exploits.

Example: how faithfulness fails — cancellation

Let $X \sim N(0, 1)$, $Z = aX + \epsilon_Z$, $Y = cX + bZ + \epsilon_Y$. Then $\mathrm{Cov}(X, Y) = c + ab$. If $c = -ab$, the direct and indirect effects cancel and $X \perp\!\!\!\perp Y$ although $X \to Y$ is an edge. The set of such coefficient values has Lebesgue measure zero, which is the usual justification of faithfulness; near-cancellation, however, makes tests unreliable in finite samples.

Example: d-separation in the health example

$A \perp E$ (the only path $A \to B \leftarrow E$ has an unconditioned collider), but $A \not\perp E \mid P$ (P is a descendant of the collider B). $E \perp R \mid B$ and $E \perp R \mid P$ (the chain is blocked). $A \perp P \mid B$.

5 Markov Equivalence

5.1 Directions with the same conditional independences cannot be distinguished

The three DAGs

$$X_1 \to X_2 \to X_3, \qquad X_1 \leftarrow X_2 \to X_3, \qquad X_1 \leftarrow X_2 \leftarrow X_3$$

all imply the same conditional independence $X_1 \perp\!\!\!\perp X_3 \mid X_2$ and nothing else. Hence from observational data alone the three directions cannot be told apart. DAGs that imply the same conditional independences form a **Markov equivalence class**.

Theorem: the three DAGs define the same Gaussian model

A Gaussian distribution on (X_1, X_2, X_3) factorizes according to one of the three DAGs iff $X_1 \perp\!\!\!\perp X_3 \mid X_2$; so all three describe exactly the same set of covariance matrices.

Proof [Supplement]. If $X_1 \perp\!\!\!\perp X_3 \mid X_2$ then $p = p(x_1)p(x_2 \mid x_1)p(x_3 \mid x_2)$ (first DAG). By Bayes' rule $p(x_1)p(x_2 \mid x_1) = p(x_2)p(x_1 \mid x_2)$, giving $p(x_2)p(x_1 \mid x_2)p(x_3 \mid x_2)$ (fork), and similarly $p(x_2 \mid x_3)p(x_3)$ replaces $p(x_2)p(x_3 \mid x_2)$ (third DAG). Conversely each factorization implies the conditional independence. Each conditional of a Gaussian is a Gaussian linear regression, so every factorization is a Gaussian linear SEM. ■

The collider $X_1 \to X_2 \leftarrow X_3$ has a different pattern, $X_1 \perp\!\!\!\perp X_3$ but $X_1 \not\perp\!\!\!\perp X_3 \mid X_2$, so it *can* be distinguished.

5.2 Skeleton and unshielded colliders

Definition: skeleton and v-structure

The **skeleton** is the adjacency structure left after deleting arrowheads: $X \to Z \to Y \mapsto X - Z - Y$. An **unshielded collider** (v-structure) is $X \to Z \leftarrow Y$ with X and Y non-adjacent.

Theorem: Verma–Pearl characterization

Two DAGs are Markov equivalent iff they have the same skeleton and the same unshielded colliders.

Idea [Supplement]. Adjacency is detectable: X, Y are adjacent iff no set d-separates them (an edge is never blocked; for non-adjacent nodes, the parents of the later one in a topological order d-separate them). Given the skeleton, an unshielded triple $X - Z - Y$ is a collider iff Z is *not* in any set separating X and Y, which again is a CI statement. Conversely, if skeletons and v-structures agree, one shows that every path is blocked by the same sets in both graphs. ■

Without further distributional assumptions, observational data usually identify not one DAG but this equivalence class.

5.3 CPDAG

Definition: CPDAG

The completed partially directed acyclic graph of an equivalence class has a directed edge $X \to Y$ if every DAG in the class contains $X \to Y$, and an undirected edge $X - Y$ if the direction varies within the class.

For instance the CPDAG $X - Z - Y$ represents $X \to Z \to Y$, $X \leftarrow Z \to Y$ and $X \leftarrow Z \leftarrow Y$ together; the collider $X \to Z \leftarrow Y$ is its own class and its CPDAG has both arrows. For the health example, the CPDAG is $A \to B \leftarrow E$, $B \to P \to R$: the v-structure at B fixes $A \to B$ and $E \to B$, and then $B \to P$ is forced (otherwise $A \to B \leftarrow P$ would be a new v-structure), and likewise $P \to R$. **An undirected edge does not mean "no relation"; it means the direction could not be distinguished.**

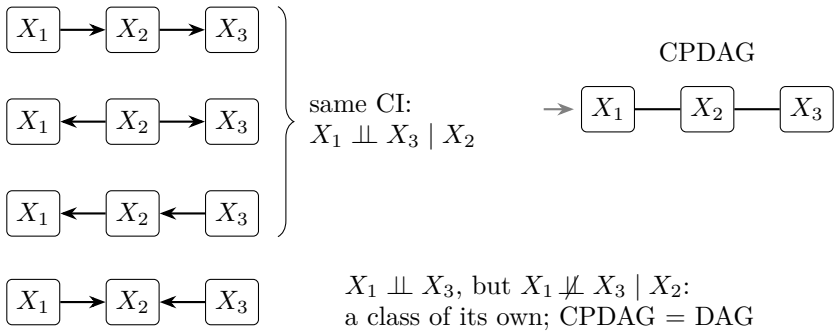


Figure 2: Markov equivalence. The chain, fork and reversed chain share the skeleton and have no v-structure, so they imply the same conditional independences; their CPDAG keeps both edges undirected. The collider has a v-structure and is identified.

Summary

- A Gaussian linear SEM writes each variable as a regression on its parents in a DAG; $\Sigma = (I - B)^{-1} D (I - B)^{-\top}$ and the GGM is the moral graph.
- Chains and forks are closed by conditioning on the middle node; colliders are opened. d-separation formalizes this.
- Under the Markov condition and faithfulness, conditional independences equal d-separations.
- Markov-equivalent DAGs share skeleton and v-structures; Gaussian observational data identify only this class, summarized by the CPDAG.

The key question of the next chapters: in Gaussian observational data, **how far can the direction of relations be inferred**, and with which algorithms?

6 Selected exercises

Motivation. The first exercise shows that each structural equation is an ordinary regression on the parents and yields the local Markov property; the second checks the rule that conditioning on a descendant of a collider opens it; the third counts the Markov equivalence class of a path.

Exercise 9.1 (Structural errors and non-descendants)

In a Gaussian linear SEM on a DAG G, let $\mathrm{nd}(j)$ be the set of non-descendants of j other than j.
(a) Show that ϵ_j is independent of $X_{\mathrm{nd}(j)}$.
(b) Deduce that the population least-squares regression of X_j on $X_{\mathrm{pa}(j)}$ has coefficients β_{kj} and residual variance σ_j^2, and that $X_j \perp\!\!\!\perp X_{\mathrm{nd}(j)\setminus\mathrm{pa}(j)} \mid X_{\mathrm{pa}(j)}$ (local Markov property).

Solution [Supplement].

(a)

Step 1. Order the variables topologically; then B is strictly lower triangular, $B^p = 0$, and $(I-B)^{-1} = \sum_{m=0}^{p-1} B^m$. The entry $(B^m)_{kl}$ is a sum over directed paths $l \to \cdots \to k$ of length m of products of edge coefficients, so $\{(I-B)^{-1}\}_{kl} \neq 0$ only if $l = k$ or l is an ancestor of k.

Step 2. If $k \in \mathrm{nd}(j)$, then $j \neq k$ and j is not an ancestor of k, so $X_k = \sum_l \{(I-B)^{-1}\}_{kl}\epsilon_l$ involves only $(\epsilon_l)_{l\neq j}$. The errors are independent, so ϵ_j is independent of $(\epsilon_l)_{l\neq j}$ and hence of $X_{\mathrm{nd}(j)}$.

(b)

Step 3. A parent of j cannot be a descendant of j (that would close a directed cycle), so $\mathrm{pa}(j) \subseteq \mathrm{nd}(j)$. Write $X_P = X_{\mathrm{pa}(j)}$ and $\beta = (\beta_{kj})_{k\in\mathrm{pa}(j)}$. Since $\mathbb{E}(X_P\epsilon_j) = 0$, β solves the normal equations $\mathbb{E}\{X_P(X_j - X_P^\top\beta)\} = 0$. These have a unique solution because $\mathrm{Var}(X_P)$ is a principal submatrix of $\Sigma = (I-B)^{-1}D(I-B)^{-\top} \succ 0$. The residual is ϵ_j, with variance σ_j^2.

Step 4. Let $X_N = X_{\mathrm{nd}(j)\setminus\mathrm{pa}(j)}$. Given $(X_P, X_N) = (x_P, x_N)$, $X_j = x_P^\top\beta + \epsilon_j$ with ϵ_j independent of (X_P, X_N) by (a), so the conditional law $N(x_P^\top\beta, \sigma_j^2)$ does not depend on x_N. This is $X_j \perp\!\!\!\perp X_N \mid X_P$. ■

Exercise 9.2 (Conditioning on a descendant of a collider)

Let $X, Y, \epsilon_Z, \epsilon_W$ be independent $N(0,1)$, $Z = aX + bY + \epsilon_Z$ and $W = cZ + \epsilon_W$ with $a, b, c \neq 0$. Show that

$$\rho_{XY\cdot W} = -\frac{c^2ab}{\sqrt{\{c^2(1+b^2)+1\}\{c^2(1+a^2)+1\}}} \neq 0,$$

and that $|\rho_{XY\cdot W}|$ is strictly smaller than the collider value $|\rho_{XY\cdot Z}| = |ab|/\sqrt{(1+a^2)(1+b^2)}$.

Solution [Supplement].

Step 1. $\mathrm{Cov}(X, W) = c\,\mathrm{Cov}(X, Z) = ca$, $\mathrm{Cov}(Y, W) = cb$, and $v := \mathrm{Var}W = c^2(a^2+b^2+1)+1$.

Step 2. The residuals of X and Y on W are $X - \frac{ca}{v}W$ and $Y - \frac{cb}{v}W$. Their covariance is $\mathrm{Cov}(X,Y) - \frac{c^2ab}{v} = -\frac{c^2ab}{v}$, and their variances are $1 - \frac{c^2a^2}{v} = \frac{c^2(1+b^2)+1}{v}$ and $\frac{c^2(1+a^2)+1}{v}$. Dividing gives the formula, which is nonzero since $a, b, c \neq 0$.

Step 3. Dividing numerator and denominator by c^2,

$$|\rho_{XY\cdot W}| = \frac{|ab|}{\sqrt{(1+b^2+c^{-2})(1+a^2+c^{-2})}} < \frac{|ab|}{\sqrt{(1+a^2)(1+b^2)}}.$$

So W opens the path $X \to Z \leftarrow Y$, as d-separation predicts, but less than Z itself; the effect approaches the collider value as $c^2 \to \infty$, when W becomes an almost noiseless copy of Z. ■

Exercise 9.3 (The equivalence class of a path)

Consider DAGs whose skeleton is the path $X_1 - X_2 - \cdots - X_p$, $p \geq 2$. Show that exactly p of them have no v-structure, namely those with a single source X_s,

$$X_1 \leftarrow \cdots \leftarrow X_{s-1} \leftarrow X_s \rightarrow X_{s+1} \rightarrow \cdots \rightarrow X_p, \qquad s = 1, \ldots, p,$$

that these form one Markov equivalence class, and that its CPDAG is the undirected path.

Solution [Supplement].

Step 1. The skeleton has no cycle, so every orientation of its edges is acyclic. Encode the edge between X_i and X_{i+1} by $e_i = \mathsf{R}$ if $X_i \rightarrow X_{i+1}$ and $e_i = \mathsf{L}$ otherwise.

Step 2. Two edges share a node only if they are consecutive. The triple (X_{i-1}, X_i, X_{i+1}) is unshielded, since X_{i-1} and X_{i+1} are not adjacent, and it is a collider iff $e_{i-1} = \mathsf{R}$ and $e_i = \mathsf{L}$. So the DAG has no v-structure iff the word $e_1 \cdots e_{p-1}$ contains no factor RL.

Step 3. If $e_i = \mathsf{R}$ and $e_j = \mathsf{L}$ for some $i < j$, let i' be the largest index below j with $e_{i'} = \mathsf{R}$; then $e_{i'+1} = \mathsf{L}$ (either $i' + 1 = j$, or $e_{i'+1} \neq \mathsf{R}$ by maximality), so RL occurs. Hence a word without RL has all its L's before all its R's: it is $\mathsf{L}^{s-1}\mathsf{R}^{p-s}$ for some $s \in \{1, \ldots, p\}$, which is exactly the DAG with single source X_s. Conversely these p words contain no RL.

Step 4. By the Verma–Pearl theorem the p DAGs are Markov equivalent (same skeleton, no v-structures), and any DAG equivalent to them has the same skeleton and no v-structure, so it is one of them. The edge e_i is R for $s \leq i$ and L for $s > i$, so its direction varies within the class, and every edge of the CPDAG is undirected. ■